

Experiment 2: C Compiler in Virtual Machine

Aim

To install the GCC C compiler inside the Ubuntu virtual machine and compile and execute a simple C program.

Tools Required

- Ubuntu VM from Experiment 1
- Ubuntu Terminal
- build-essential / GCC

Detailed Procedure

Open Terminal in Ubuntu using Ctrl + Alt + T.

```
sudo apt update
```

Install the GCC compiler and supporting build tools.

```
sudo apt install build-essential -y
```

Verify the compiler installation.

```
gcc --version
```

Create the C source file.

```
nano hello.c
```

Enter the following program in nano:

```
#include <stdio.h>

int main() {
    printf("Hello, Virtual Cloud World!\n");
    return 0;
}
```

Save the file and exit nano.

```
Ctrl + O → Enter
Ctrl + X
```

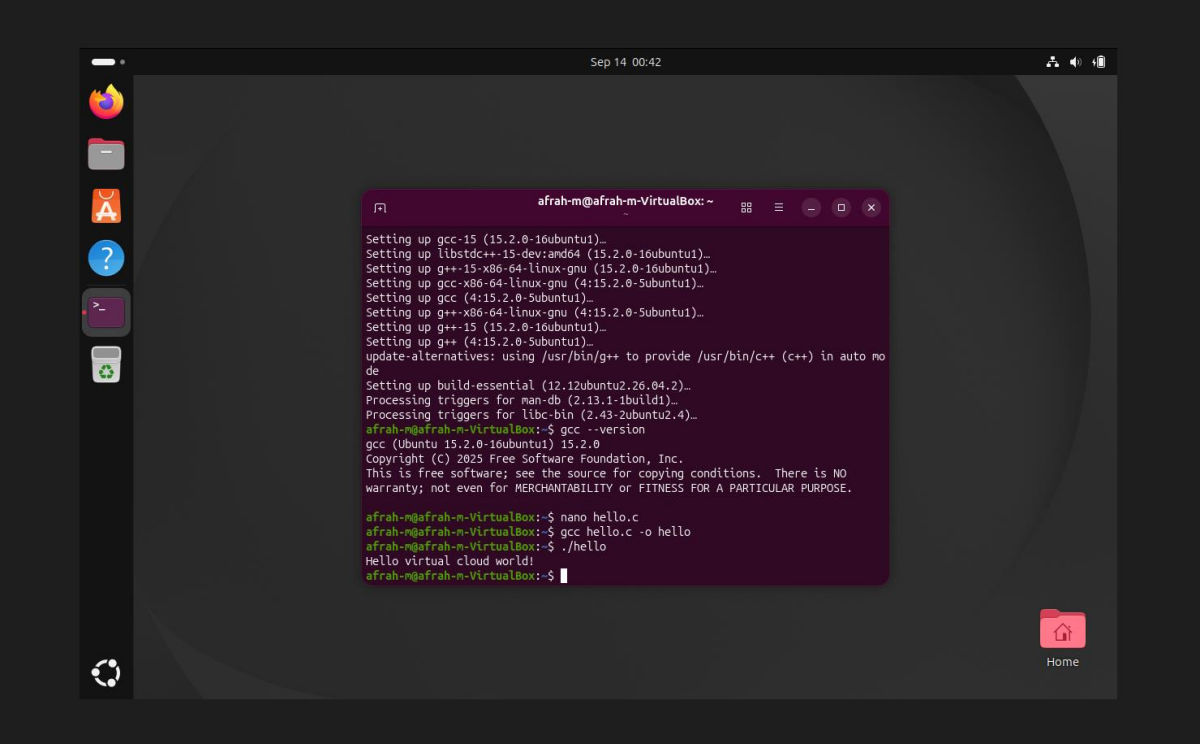
Compile the program and create an executable named hello.

```
gcc hello.c -o hello
```

Execute the compiled program.

```
./hello
```

Output



The screenshot shows a terminal window titled 'afrah-m@afrah-m-VirtualBox: ~' with a dark purple background. The terminal output displays the installation of GCC 15 and related dependencies. The user then creates a C file named 'hello.c', compiles it with 'gcc hello.c -o hello', and runs the executable with './hello', which outputs 'Hello virtual cloud world!'. The terminal window is open on a desktop environment with a dark theme, a sidebar on the left containing application icons, and a 'Home' folder icon on the right.

```
afrah-m@afrah-m-VirtualBox: ~  
Setting up gcc-15 (15.2.0-16ubuntu1)...  
Setting up libstdc++15-dev:amd64 (15.2.0-16ubuntu1)...  
Setting up g++-15-x86-64-linux-gnu (15.2.0-16ubuntu1)...  
Setting up gcc-x86-64-linux-gnu (4:15.2.0-5ubuntu1)...  
Setting up gcc (4:15.2.0-5ubuntu1)...  
Setting up g++-x86-64-linux-gnu (4:15.2.0-5ubuntu1)...  
Setting up g++-15 (15.2.0-16ubuntu1)...  
Setting up g++ (4:15.2.0-5ubuntu1)...  
update-alternatives: using /usr/bin/g++ to provide /usr/bin/c++ (c++) in auto mode  
Setting up build-essential (12.12ubuntu2.26.04.2)...  
Processing triggers for man-db (2.13.1-1build1)...  
Processing triggers for libc-bin (2.43-2ubuntu2.4)...  
afrah-m@afrah-m-VirtualBox:~$ gcc --version  
gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0  
Copyright (C) 2025 Free Software Foundation, Inc.  
This is free software; see the source for copying conditions. There is NO  
warranty; not even for MERCHANTABILITY or FITNESS FOR A PARTICULAR PURPOSE.  
  
afrah-m@afrah-m-VirtualBox:~$ nano hello.c  
afrah-m@afrah-m-VirtualBox:~$ gcc hello.c -o hello  
afrah-m@afrah-m-VirtualBox:~$ ./hello  
Hello virtual cloud world!  
afrah-m@afrah-m-VirtualBox:~$
```

Result

GCC was installed successfully, the C program was compiled without errors, and the executable produced the expected output.